

UNIFORMLY TIGHT ANCIENT NAVIER–STOKES DYNAMICS AND CONDITIONAL EQUILIBRIUM RIGIDITY

GUILLEM DURAN BALLESTER

ABSTRACT. We study bounded ancient solutions of the three-dimensional Navier–Stokes equations in backward self-similar variables under uniform L^3 -tightness. We first establish the analytic compactness framework: bounded mild ancient solutions are smooth, locally compact under time translation, and the uniform $L^\infty_\tau L^3_y$ and tightness bounds are stable under locally smooth limits. We then prove a minimal-element reduction for closed normalized uniformly tight collections: any nonempty such collection contains a nonzero solution of minimal $L^\infty(\mathbb{R}^3 \times \mathbb{R})$ size.

For the time-translation hull of a bounded ancient solution we construct compact invariant hulls and invariant probability measures by Krylov–Bogolyubov averaging. These measures satisfy a stationary statistical form of the self-similar Navier–Stokes equation. Passing to the barycenter produces an explicit quadratic covariance term, isolating the obstruction to obtaining a single stationary self-similar profile.

Finally, we formulate a strict compact-hull Lyapunov hypothesis. Under this hypothesis, invariant measures are supported on stationary trajectories. Hence a minimal compact uniformly L^3 -bounded hull consists of a single stationary profile, which is zero by the stationary self-similar theorem of Nečas–Růžička–Šverák. As a consequence, the uniformly L^3 -tight branch of the Type I ancient-profile reduction is excluded provided the corresponding nonzero compact-recurrence and generated Lyapunov properties hold.

1. INTRODUCTION AND MAIN RESULTS

1.1. The uniformly tight ancient branch. We consider the three-dimensional incompressible Navier–Stokes equations in backward self-similar variables. The centered velocity V and pressure P solve

$$(1.1) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$. In the companion Type I reduction paper [2], admissible Type I singularities are reduced to nonzero normalized ancient solutions of (1.1). The present paper starts after that reduction and studies the branch in which

$$(1.2) \quad \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$(1.3) \quad \lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

We call this the uniformly L^3 -tight ancient branch.

This profile condition is distinct from the source-level condition

$$u \in L^\infty(0, T; L^3(\mathbb{R}^3)).$$

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For finite-energy suitable weak solutions, the source-level assumption $u \in L_t^\infty L_x^3$ near a putative singular time is incompatible with singularity by the theorem of Escauriaza–Seregin–Šverák [3]. Uniform L^3 -tightness in this paper is a condition on centered ancient profiles after blow-up.

1.2. Minimal elements and compact hulls. The first reduction is variational:

$$\text{nonempty tight normalized class} \implies L^\infty\text{-minimal ancient element.}$$

The normalization is a fixed local velocity nonvanishing condition. It prevents minimizing sequences from losing all mass by time translation. Once a minimal element V_* is selected, its time-translation hull is

$$\mathcal{H}(V_*) = \overline{\{\Theta_s V_* : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}, \quad (\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The hull is compact because bounded mild ancient solutions are locally smooth with estimates depending only on their common L^∞ bound.

1.3. Invariant measures and covariance. Every compact invariant hull K carries invariant probability measures. Indeed, Krylov–Bogolyubov averages

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds$$

have weak subsequential limits. Such invariant measures satisfy a stationary statistical form of (1.1). If

$$\overline{W}(y) = \int_K W(y, 0) d\mu(W),$$

then the averaged nonlinearity decomposes as

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Thus the barycenter solves the stationary self-similar equation only up to the explicit covariance defect C . The invariant-measure argument alone does not prove that a recurrent compact hull contains a single stationary profile.

1.4. The strict Lyapunov hypothesis. The present paper therefore isolates the exact conditional input needed to pass from compact recurrent dynamics to equilibria. It does not construct a universal Lyapunov functional for all compact ancient Navier–Stokes hulls. Instead, it proves that if such a strict Lyapunov functional exists on the compact hull generated by a minimal tight element, then the tight branch is excluded.

The logical status of the paper is summarized as follows.

Step	Status in this paper
bounded ancient smoothing	proved
local smooth compactness	proved
closure of uniform L^3 -tightness	proved
minimal L^∞ element	proved
compact trajectory hull	proved
invariant measures	proved
stationary statistical identity	proved
covariance obstruction	identified
strict Lyapunov functional	assumed
nonzero compact recurrence of minimal element	assumed
compact tight rigidity under Lyapunov	proved
tight-class input for the companion paper	conditional corollary

Thus the paper does not prove an unconditional uniformly tight Liouville theorem. It proves that the uniformly tight branch is empty once the two compact-dynamical inputs RC and LY formulated in [Theorems 9.1](#) and [9.2](#) are available.

1.5. Main results. The first headline theorem is the analytic compactness and closure statement.

Theorem 1.1 (Compactness and closure of uniformly tight ancient solutions). *Let $\{V_n\}$ be a sequence of bounded mild ancient solutions of [\(1.1\)](#). Assume*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

If in addition there are $L < \infty$ and a modulus $\omega : [1, \infty) \rightarrow [0, \infty)$, $\omega(R) \rightarrow 0$, such that

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R) \quad (R \geq 1),$$

then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L, \quad \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R),$$

and in particular V is uniformly L^3 -tight.

Compatible pressure representatives also converge locally weakly modulo time-dependent gauges under the additional $L^\infty_\tau L^3_y$ assumption; see [Theorem 3.3](#).

The second headline theorem is the minimal-element reduction.

Theorem 1.2 (Minimal element in a normalized uniformly tight class). *Let A be a nonempty collection of bounded mild ancient solutions satisfying a common L^∞ bound, a common $L^\infty_\tau L^3_y$ bound, a common uniform L^3 -tightness modulus, and a fixed local nonvanishing normalization*

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 > 0.$$

Assume A is invariant under time translation and closed under locally smooth limits which retain the normalization. Then A contains an element V_ minimizing*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}$$

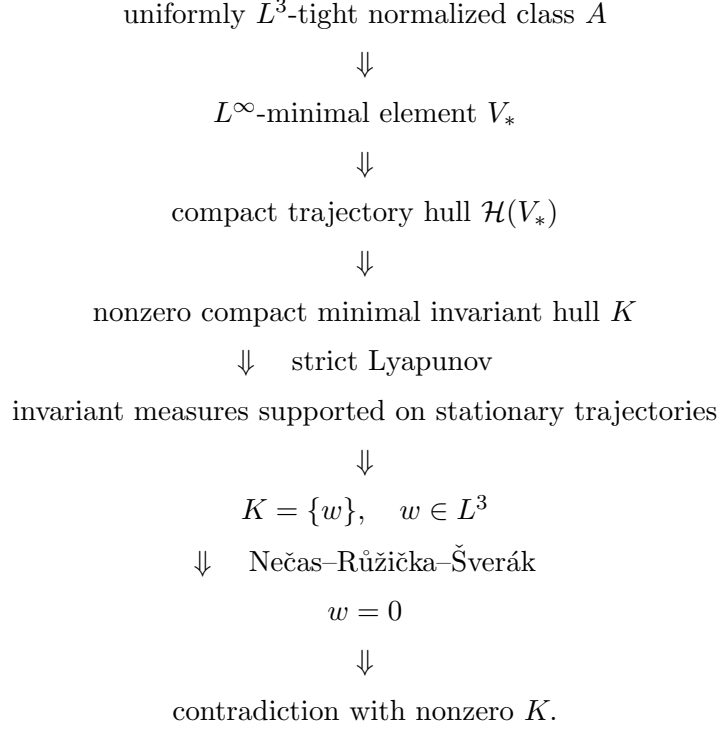
among all $V \in A$. In particular $V_ \not\equiv 0$.*

The third headline theorem is the conditional tight-branch Liouville criterion.

Theorem 1.3 (Conditional uniformly tight Liouville criterion). *Let A be a normalized uniformly L^3 -tight collection as in [Theorem 1.2](#). If A satisfies the recurrence property RC(A) and the generated Lyapunov property LY(A) of [Theorems 9.1](#) and [9.2](#), then*

$$A = \emptyset.$$

The proof of the last theorem follows the dependency diagram



1.6. Relation to the companion paper. The companion paper [2] reduces admissible Type I singularities to normalized Seregin ancient limits and organizes such limits into small, stationary L^3 , uniformly tight, structured/decay, and residual classes. The present paper supplies the conditional tight-class input: under the compact-recurrence and strict Lyapunov properties stated below, the uniformly L^3 -tight class contains no nonzero normalized ancient limit.

1.7. Organization of the paper. Section 2 fixes the centered equation, mild formulation, time translation, and local smooth topology. Section 3 proves smoothing, pressure reconstruction, stability, and compactness. Section 4 proves closure of uniform L^3 -tightness and recalls stationary L^3 rigidity. Section 5 proves the minimal-element reduction. Section 6 constructs compact trajectory hulls and minimal invariant subsets. Section 7 constructs invariant measures and identifies the covariance obstruction. Section 8 proves the consequences of the strict Lyapunov hypothesis. Section 9 assembles the conditional tight-branch Liouville criterion and records the interface with the companion residual-criterion paper.

2. CENTERED ANCIENT SOLUTIONS AND THE LOCAL SMOOTH TOPOLOGY

2.1. The centered equation. Throughout the paper V denotes a centered ancient velocity and P its pressure. A generic trajectory in a compact hull is denoted by W , and a stationary profile by $w(y)$. Define

$$LV = \Delta V - \frac{1}{2}y \cdot \nabla V - \frac{1}{2}V.$$

After applying the Helmholtz–Leray projection $\mathbb{P}$, the centered equation becomes

$$(2.1) \quad \partial_\tau V = LV - \mathbb{P} \operatorname{div}(V \otimes V).$$

2.2. The Ornstein–Uhlenbeck–Stokes semigroup. Let $S(a) = e^{aL}$, $a > 0$. It is given by

$$(2.2) \quad (S(a)f)(y) = e^{-a/2} \int_{\mathbb{R}^3} \frac{\exp\left(-\frac{|z - e^{-a/2}y|^2}{4(1 - e^{-a})}\right)}{[4\pi(1 - e^{-a})]^{3/2}} f(z) dz.$$

The factor $e^{-a/2}$ records the damping caused by the term $-V/2$.

Lemma 2.1 (Centered Stokes kernel estimate). *For $a > 0$ and $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(2.3) \quad \|(S(a)\mathbb{P} \operatorname{div})^* \phi\|_{L^1(\mathbb{R}^3)} \leq C e^{-a/2} (1 - e^{-a})^{-1/2} \|\phi\|_{L^1(\mathbb{R}^3)}.$$

Consequently the Duhamel term in (2.4) is well-defined by duality for bounded V .

Proof. Set $\lambda = e^{-a/2}$ and $\theta = 1 - e^{-a}$. From (2.2),

$$S(a)f(y) = e^{-a/2} (e^{\theta\Delta} f)(\lambda y).$$

Thus

$$S(a)^* \phi(z) = e^{-a/2} \lambda^{-3} (e^{\theta\Delta} \psi)(z), \quad \psi(x) = \phi(x/\lambda),$$

and $\|\psi\|_{L^1} = \lambda^3 \|\phi\|_{L^1}$. The Leray projection is a homogeneous Fourier multiplier of degree zero, so it commutes with this dilation. The classical heat–Stokes kernel estimate

$$\|(e^{\theta\Delta} \mathbb{P} \operatorname{div})^* \phi\|_{L^1} \leq C \theta^{-1/2} \|\phi\|_{L^1}, \quad \theta > 0,$$

follows from the Oseen kernel bound; see [4, 5, 10]. Substituting $\theta = 1 - e^{-a}$ and using

$$e^{-a/2} \lambda^{-3} \|\psi\|_{L^1} = e^{-a/2} \|\phi\|_{L^1}$$

gives (2.3). Since $(1 - e^{-a})^{-1/2}$ is integrable at $a = 0$, the duality pairing in (2.4) is integrable on finite time intervals for $V \in L^\infty$. $\square$

2.3. Riesz transform convention. We use the convention that R_i has Fourier symbol $i\xi_i/|\xi|$. Thus $R_i R_j$ has symbol $-\xi_i \xi_j / |\xi|^2$, and

$$P = R_i R_j (F_{ij}) \iff -\Delta P = \partial_i \partial_j F_{ij}.$$

Pressures are determined only modulo functions of time unless a whole-space Leray gauge is explicitly chosen.

2.4. Bounded mild ancient solutions.

Definition 2.2 (Bounded mild ancient solution). A vector field $V : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$ is a bounded mild ancient solution of (1.1) if

$$V \in L^\infty(\mathbb{R}^3 \times \mathbb{R}), \quad \operatorname{div} V = 0$$

in distributions, and for every $\sigma < \tau$,

$$(2.4) \quad V(\tau) = S(\tau - \sigma)V(\sigma) - \int_\sigma^\tau S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s) ds$$

in the weak-* topology of $L^\infty(\mathbb{R}^3)$.

The Duhamel term in (2.4) is interpreted by duality. For $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V \otimes V)(s), \phi \rangle = \langle V \otimes V(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle,$$

and Theorem 2.1 makes the scalar integrand absolutely integrable on $s \in (\sigma, \tau)$.

2.5. Time translation. For $s \in \mathbb{R}$ define

$$(\Theta_s V)(y, \tau) = V(y, \tau + s).$$

The equation, the mild formulation, the L^∞ norm, and the $L^\infty_\tau L^3_y$ and tail bounds used below are invariant under Θ_s . Indeed, replacing (σ, τ) by $(\sigma + s, \tau + s)$ in (2.4) and changing variables in the time integral gives the mild formula for $\Theta_s V$.

2.6. The locally smooth topology.

Definition 2.3 (Local smooth topology). The locally smooth topology on $C^\infty_{\text{loc}}(\mathbb{R}^3 \times \mathbb{R}; \mathbb{R}^3)$ is generated by the seminorms

$$\|V\|_{m,N} = \max_{|\alpha|+j \leq m} \sup_{\substack{|y| \leq N \\ |\tau| \leq N}} |\partial_y^\alpha \partial_\tau^j V(y, \tau)|, \quad m, N \in \mathbb{N}.$$

One compatible metric is

$$d(U, V) = \sum_{m,N=1}^{\infty} 2^{-(m+N)} \frac{\|U - V\|_{m,N}}{1 + \|U - V\|_{m,N}}.$$

Thus $V_n \rightarrow V$ locally smoothly if and only if the convergence is in $C^m(K)$ for every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and every $m \geq 0$.

3. SMOOTHING, PRESSURE RECONSTRUCTION, AND COMPACTNESS

3.1. Local smoothing.

Lemma 3.1 (Smoothing of bounded mild solutions). *Every bounded mild ancient solution is smooth on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies the projected equation (2.1) classically. Moreover, for each compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ and each integer $m \geq 0$,*

$$\|V\|_{C^m(K)} \leq C(m, K, \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}).$$

The same estimate holds uniformly for any family with a common L^∞ bound.

Proof. Fix a compact time interval $J \Subset \mathbb{R}$ containing the time projection of K , and set $t = -e^{-\tau}$. On the corresponding compact interval $t(J) \Subset (-\infty, 0)$, define

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)).$$

The map $(x, t) \mapsto (x/\sqrt{-t}, -\log(-t))$ is a smooth diffeomorphism between $\mathbb{R}^3 \times t(J)$ and $\mathbb{R}^3 \times J$, with all derivatives bounded on compact subcylinders.

The centered mild formula (2.4) transforms, under this parabolic change of variables, into the usual projected Navier–Stokes mild formula on every compact subinterval of $t(J)$:

$$U(t) = e^{(t-s)\Delta} U(s) - \int_s^t e^{(t-r)\Delta} \mathbb{P} \operatorname{div}(U \otimes U)(r) dr, \quad s < t.$$

This transformation uses only the projected mild equation: the Leray projection is homogeneous of degree zero and commutes with the parabolic dilation, while the Ornstein–Uhlenbeck kernel in (2.2) becomes the heat kernel with time parameter $t - s$. No pressure formulation is used at this stage. The L^∞ norm of U on $\mathbb{R}^3 \times t(J)$ is bounded by a constant depending only on J and $\|V\|_{L^\infty}$.

The standard local smoothing theorem for bounded projected mild Navier–Stokes solutions [6, 5] gives first local derivative bounds away from the initial time in any mild representation. In the present ancient setting one may choose the initial time strictly before the compact subcylinder under consideration. Iterating the differentiated Duhamel formula and then

applying the interior parabolic bootstrap for the Stokes system [8, 10] gives uniform bounds for all derivatives of U on compact subcylinders of $\mathbb{R}^3 \times t(J)$. The mild formula may then be differentiated in time, so U solves the projected Navier–Stokes equation classically. Returning to (y, τ) gives the asserted estimates for V and the classical projected equation (2.1). The argument is uniform for families with a common L^∞ bound. $\square$

3.2. Pressure reconstruction.

Lemma 3.2 (Local pressure reconstruction). *Let V be a smooth bounded solution of the projected centered equation (2.1). Then for every ball B_R and compact interval $I \subseteq \mathbb{R}$ there exists*

$$P_R \in C^\infty(B_R \times I),$$

unique up to an additive function of time, such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P_R = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \operatorname{div} V = 0,$$

on $B_R \times I$.

Proof. Set

$$F = \Delta V - \frac{1}{2}(V + y \cdot \nabla V) - \partial_\tau V - (V \cdot \nabla)V.$$

Since $(V \cdot \nabla)V = \operatorname{div}(V \otimes V)$ and V satisfies (2.1), self-adjointness of the Leray projection gives

$$\int_I \int_{B_R} F \cdot \Phi \, dy \, d\tau = 0$$

for every $\Phi \in C_c^\infty(B_R \times I; \mathbb{R}^3)$ satisfying $\operatorname{div}_y \Phi = 0$. The local de Rham theorem, applied in the spatial variables with τ as a parameter, gives a scalar distribution P_R on $B_R \times I$, unique up to a distribution depending only on time, such that $\nabla_y P_R = F$. Since F is smooth, all spatial derivatives of P_R of positive order are smooth. Fixing, for instance, the spatial mean of P_R over B_R to be zero for each τ selects a representative which is smooth on $B_R \times I$. This proves the local pressure formulation. $\square$

Lemma 3.3 (Leray pressure and local gauges). *Let V be a bounded mild ancient solution satisfying*

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then the whole-space Leray pressure

$$P = R_i R_j (V_i V_j)$$

belongs to $L^\infty(\mathbb{R}; L^{3/2}(\mathbb{R}^3))$. Moreover, if $V_n \rightarrow V$ locally smoothly, the V_n have a common $L_\tau^\infty L_y^3$ bound, and

$$P_n = R_i R_j (V_{n,i} V_{n,j}),$$

then on every compact cylinder $B_R \times [\tau_1, \tau_2]$ there are bounded measurable gauges $a_n(\tau)$ and $a(\tau)$ such that, after passing to a subsequence,

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

Equivalently, after replacing the limiting pressure by an equivalent representative modulo a function of time, the weak limit may be denoted by P .

Proof. The first assertion follows from the Calderon–Zygmund boundedness of Riesz transforms on $L^{3/2}(\mathbb{R}^3)$ [11]:

$$\|P(\tau)\|_{L^{3/2}} \leq C\|V_i(\tau)V_j(\tau)\|_{L^{3/2}} \leq C\|V(\tau)\|_{L^3}^2.$$

For the local stability statement, choose $\chi \in C_c^\infty(B_{4R})$ with $\chi \equiv 1$ on B_{3R} . Decompose on each time slice

$$P_n = R_i R_j (\chi V_{n,i} V_{n,j}) + R_i R_j ((1 - \chi) V_{n,i} V_{n,j}).$$

The local term converges strongly in $L^q(B_{2R} \times [\tau_1, \tau_2])$ for every finite q . This is because $V_n \rightarrow V$ smoothly on $\bar{B}_{4R} \times [\tau_1, \tau_2]$, so $\chi V_{n,i} V_{n,j} \rightarrow \chi V_i V_j$ strongly in every finite L^q , and the Riesz transforms are Calderon–Zygmund operators on L^q , $1 < q < \infty$.

It remains to control the exterior part modulo constants. Let K_{ij} be the convolution kernel of $R_i R_j$. For $x \in B_R$ and $|z| > 4R$,

$$|K_{ij}(x - z) - K_{ij}(-z)| \leq C_R |z|^{-4}.$$

Define the exterior gauges

$$b_n(\tau) = \int_{\mathbb{R}^3} K_{ij}(-z)(1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz$$

and $b(\tau)$ by the same formula with V in place of V_n . The cutoff removes the singularity at the origin, and the integral is absolutely convergent at infinity because, on the dyadic annulus $A_m = \{2^m R < |z| < 2^{m+1} R\}$,

$$\int_{A_m} |z|^{-3} |V_n(z, \tau)|^2 dz \leq C(2^m R)^{-3} \|V_n(\tau)\|_{L^3(A_m)}^2 |A_m|^{1/3} \leq CL^2(2^m R)^{-2}.$$

The same dyadic estimate shows that b_n and b are bounded uniformly on $[\tau_1, \tau_2]$. The maps $\tau \mapsto b_n(\tau)$ and $\tau \mapsto b(\tau)$ are measurable, because they are limits of measurable truncated integrals. After subtracting $b_n(\tau)$, the exterior contribution on B_R is

$$E_n(x, \tau) = \int_{\mathbb{R}^3} [K_{ij}(x - z) - K_{ij}(-z)](1 - \chi(z))V_{n,i}(z, \tau)V_{n,j}(z, \tau) dz.$$

The same dyadic estimate, now using $|z|^{-4}$, gives

$$\|E_n(\cdot, \tau)\|_{L^\infty(B_R)} \leq C(R)L^2$$

uniformly in n and τ . Splitting the exterior integral into $\{4R < |z| < R'\}$ and $\{|z| > R'\}$, the first part converges by local smooth convergence, while the second part is bounded uniformly by CL^2/R' . Letting first $n \rightarrow \infty$ and then $R' \rightarrow \infty$ identifies the local limit of E_n with the corresponding exterior contribution for V . Therefore, with $a_n = b_n$ and $a = b$, the decompositions of $P_n - a_n$ and $P - a$ have local parts converging strongly and exterior parts converging distributionally with a uniform local $L^{3/2}$ bound. Hence

$$P_n - a_n(\tau) \rightharpoonup P - a(\tau) \quad \text{weakly in } L^{3/2}(B_R \times [\tau_1, \tau_2]).$$

□

3.3. Stability under locally smooth limits.

Lemma 3.4 (Closure of the mild formulation). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

If $V_n \rightarrow V$ locally smoothly, then V is a bounded mild ancient solution of (1.1).

Proof. The common L^∞ bound passes to V by pointwise convergence on compact subsets. Divergence freeness passes distributionally. Fix $\sigma < \tau$ and test (2.4) for V_n against $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$. The endpoint terms are

$$\langle V_n(\tau), \phi \rangle, \quad \langle V_n(\sigma), S(\tau - \sigma)^* \phi \rangle.$$

The first converges by local smooth convergence. For the second, note that $S(\tau - \sigma)^* \phi \in L^1(\mathbb{R}^3)$. Approximate this L^1 function by a compactly supported smooth function and use the common L^∞ bound to control the L^1 approximation error uniformly in n .

For the Duhamel term, write

$$\langle S(\tau - s) \mathbb{P} \operatorname{div}(V_n \otimes V_n)(s), \phi \rangle = \langle V_n(s) \otimes V_n(s), (S(\tau - s) \mathbb{P} \operatorname{div})^* \phi \rangle.$$

For fixed s , the adjoint test belongs to L^1 ; pairing convergence again follows by compact approximation in L^1 , local smooth convergence, and the common L^∞ bound on $V_n \otimes V_n$. The approximation error is uniform in n for this fixed s . Moreover Theorem 2.1 gives the integrable domination

$$C \sup_n \|V_n\|_{L^\infty}^2 e^{-(\tau-s)/2} (1 - e^{-(\tau-s)})^{-1/2} \|\phi\|_{L^1}, \quad s \in (\sigma, \tau).$$

Dominated convergence in s passes the Duhamel integral to the limit. Thus V satisfies (2.4). $\square$

3.4. Compactness of bounded families.

Proposition 3.5 (Compactness of bounded ancient families). *Let $\{V_n\}$ be bounded mild ancient solutions satisfying*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty.$$

Then a subsequence converges locally smoothly to a bounded mild ancient solution V .

Proof. By Theorem 3.1, the sequence is bounded in every C^m -norm on every compact cylinder. Arzela–Ascoli on a countable exhaustion $B_N \times [-N, N]$, followed by a diagonal extraction over (m, N) , gives a subsequence converging locally smoothly to a smooth limit. Theorem 3.4 shows that the limit is again a bounded mild ancient solution. $\square$

4. UNIFORM L^3 -TIGHTNESS AND STATIONARY L^3 RIGIDITY

4.1. Uniform L^3 -tightness.

Definition 4.1 (Uniform L^3 -tightness). A bounded ancient solution V is uniformly L^3 -tight if

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty$$

and

$$\lim_{R \rightarrow \infty} \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy = 0.$$

4.2. Closedness under locally smooth convergence.

Lemma 4.2 (Closure of uniform L^3 -tightness). *Let V_n be bounded mild ancient solutions satisfying*

$$\sup_n \sup_{\tau \in \mathbb{R}} \|V_n(\tau)\|_{L^3(\mathbb{R}^3)} \leq L < \infty$$

and assume that they have a common tightness modulus: for some $\omega : [1, \infty) \rightarrow [0, \infty)$ with $\omega(R) \rightarrow 0$,

$$\sup_n \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V_n(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

If $V_n \rightarrow V$ locally smoothly, then

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

In particular V is uniformly L^3 -tight.

Proof. Fix τ . On B_R , local smooth convergence gives

$$\int_{B_R} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{B_R} |V_n(y, \tau)|^3 dy \leq L^3.$$

Letting $R \rightarrow \infty$ and using monotone convergence gives $\|V(\tau)\|_{L^3} \leq L$. Since τ was arbitrary, the same bound is uniform in time.

For the tail bound, fix $1 \leq R < R'$. On the bounded annulus $\{R < |y| < R'\}$, dominated convergence gives

$$\int_{R < |y| < R'} |V(y, \tau)|^3 dy = \lim_{n \rightarrow \infty} \int_{R < |y| < R'} |V_n(y, \tau)|^3 dy \leq \omega(R).$$

Letting $R' \rightarrow \infty$ and using monotone convergence gives

$$\int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R).$$

The estimate is independent of τ , so taking the supremum in time gives the asserted tail bound. Since $\omega(R) \rightarrow 0$, V is uniformly L^3 -tight. $\square$

Proof of Theorem 1.1. The compactness assertion is Theorem 3.5. The $L_\tau^\infty L_y^3$ and tightness closure assertions follow from Theorem 4.2 with the displayed constants L and ω . $\square$

4.3. Stationary time-translate limits. The stationary self-similar profile equation used below is

$$(4.1) \quad (w \cdot \nabla)w + \nabla p = \Delta w - \frac{1}{2}(w + y \cdot \nabla w), \quad \operatorname{div} w = 0.$$

Theorem 4.3 (Stationary limits under uniform L^3 -tightness). *Let V be uniformly L^3 -tight. Let $\tau_n \in \mathbb{R}$, and suppose*

$$\Theta_{\tau_n} V \rightarrow W$$

locally smoothly. If W is stationary, then $W \equiv 0$.

Proof. The sequence $\Theta_{\tau_n} V$ has the same L^3 bound and tightness modulus as V . By Theorem 4.2, the limit W belongs to $L_\tau^\infty L_y^3$. If W is stationary, write $W(y, \tau) = w(y)$. Then $w \in L^3(\mathbb{R}^3)$.

By Theorem 3.4, W is a bounded mild ancient solution. The smoothness from Theorem 3.1 and the local pressure lemma Theorem 3.2 give smooth local pressures p_N on B_N at time 0. Since $\partial_\tau W = 0$, each p_N satisfies

$$(w \cdot \nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w) \quad \text{on } B_N.$$

On overlaps the gradients of the local pressures agree, so the pressures differ only by constants. Adjusting those constants and passing to the compatible union gives a scalar distribution p on $\mathbb{R}^3$ for which (4.1) holds in distributions. The stationary L^3 rigidity theorem Theorem 4.4 applies and gives $w = 0$. $\square$

4.4. Stationary self-similar rigidity.

Theorem 4.4 (Stationary self-similar rigidity). *Let $w \in C^\infty(\mathbb{R}^3; \mathbb{R}^3) \cap L^3(\mathbb{R}^3)$ solve (4.1) in distributions for some scalar distribution p , with $\operatorname{div} w = 0$. Then $w \equiv 0$.*

Proof. This is the theorem of Nečas, Růžička, and Šverák [9], applied to the stationary Leray self-similar system. The smooth distributional formulation stated here is a special case of their weak $L^3(\mathbb{R}^3)$ theorem. $\square$

5. NORMALIZED COLLECTIONS AND MINIMAL ANCIENT SOLUTIONS

5.1. Closed normalized collections.

Definition 5.1 (Closed normalized collection). A collection A of bounded mild ancient solutions is called a closed normalized collection if:

(i) there is $M < \infty$ such that

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M \quad \text{for all } V \in A;$$

(ii) there are $R_0 > 0$ and $\eta_0 > 0$ such that

$$(5.1) \quad \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0 \quad \text{for all } V \in A;$$

(iii) A is invariant under time translation:

$$V \in A, s \in \mathbb{R} \implies \Theta_s V \in A;$$

(iv) if $V_n \in A$, $V_n \rightarrow V$ locally smoothly, and V satisfies the same (R_0, η_0) -nonvanishing condition (5.1), then $V \in A$.

5.2. Time-centering.

Lemma 5.2 (Time-centering). *Let A be a closed normalized collection. If $V \in A$, then for every $\delta > 0$ there exists $s \in \mathbb{R}$ such that*

$$\int_{B_{R_0}} |\Theta_s V(y, 0)|^3 dy \geq \eta_0 - \delta.$$

Proof. By (5.1), the supremum in time of the local L^3 mass is at least η_0 . Choose s within δ of this supremum. Since $(\Theta_s V)(y, 0) = V(y, s)$, the displayed estimate follows. Time-translation invariance keeps $\Theta_s V$ in A . $\square$

5.3. Stability of local nonvanishing.

Lemma 5.3 (Stability of local nonvanishing). *If $V_n \rightarrow V$ locally smoothly and*

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - o(1),$$

then

$$\int_{B_{R_0}} |V(y, 0)|^3 dy \geq \eta_0.$$

Proof. Local smooth convergence implies uniform convergence on $\overline{B_{R_0}} \times \{0\}$, hence convergence in $L^3(B_{R_0})$. The map $f \mapsto \int_{B_{R_0}} |f|^3$ is continuous on $L^3(B_{R_0})$, and the claim follows by passing to the limit. $\square$

5.4. Lower semicontinuity of size.

Lemma 5.4 (Lower semicontinuity of L^∞ size). *If $V_n \rightarrow V$ locally smoothly and*

$$\sup_n \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} < \infty,$$

then

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Let L_0 be the liminf on the right. If $|V(y_0, \tau_0)| > L_0 + \varepsilon$, choose a subsequence with $\|V_n\|_{L^\infty} \rightarrow L_0$. Local uniform convergence at (y_0, τ_0) gives $|V_n(y_0, \tau_0)| > L_0 + \varepsilon/2$ for all large n , contradicting $\|V_n\|_{L^\infty} \rightarrow L_0$. Hence $|V| \leq L_0$ pointwise. Since the smooth representative has pointwise supremum equal to its essential L^∞ norm, the result follows. $\square$

5.5. Existence of a minimal element.

Theorem 5.5 (Existence of a minimal ancient solution). *Let A be nonempty and closed normalized. Set*

$$m = \inf_{V \in A} \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Then $0 < m < \infty$, and there exists $V_ \in A$ such that*

$$\|V_*\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} = m.$$

In particular $V_ \not\equiv 0$.*

Proof. The common L^∞ bound gives $m < \infty$. The local nonvanishing gives a positive lower bound:

$$\eta_0 \leq \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \leq |B_{R_0}| \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}^3,$$

so

$$m \geq (\eta_0/|B_{R_0}|)^{1/3} > 0.$$

Choose $V_n \in A$ with

$$m \leq \|V_n\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq m + \frac{1}{n}.$$

By [Theorem 5.2](#), after replacing V_n by a time translate, which stays in A and preserves the L^∞ norm, we may assume

$$\int_{B_{R_0}} |V_n(y, 0)|^3 dy \geq \eta_0 - \frac{1}{n}.$$

The common L^∞ bound and [Theorem 3.5](#) give a locally smooth subsequential limit V_* , which is a bounded mild ancient solution. By [Theorem 5.3](#),

$$\int_{B_{R_0}} |V_*(y, 0)|^3 dy \geq \eta_0,$$

so V_* retains the fixed (R_0, η_0) -normalization. Closedness of A then gives $V_* \in A$. Finally, [Theorem 5.4](#) gives

$$\|V_*\|_{L^\infty} \leq \liminf_{n \rightarrow \infty} \|V_n\|_{L^\infty} = m.$$

Since m is the infimum over A and $V_* \in A$, equality holds. The positive lower bound on m rules out $V_* \equiv 0$. $\square$

5.6. The uniformly tight normalized class.

Definition 5.6 (Uniformly tight normalized collection). Fix $M, L, R_0, \eta_0 > 0$ and a modulus

$$\omega : [1, \infty) \rightarrow [0, \infty), \quad \omega(R) \rightarrow 0.$$

A collection A of bounded mild ancient solutions is called a uniformly tight normalized collection with data $(M, L, \omega, R_0, \eta_0)$ if:

(1) for every $V \in A$,

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M;$$

(2) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq L;$$

(3) for every $V \in A$ and every $R \geq 1$,

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy \leq \omega(R);$$

(4) for every $V \in A$,

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0;$$

(5) A is invariant under time translation;

(6) A is closed under locally smooth limits retaining item (4).

Proposition 5.7 (Uniformly tight normalized collections are closed normalized). *Every uniformly tight normalized collection is a closed normalized collection.*

Proof. The L^∞ bound, local nonvanishing, time-translation invariance, and closedness clause in [Theorem 5.1](#) are part of [Theorem 5.6](#). The only non-formal point is that the stated L^3 bound and fixed tightness modulus survive locally smooth limits. This follows from [Theorem 4.2](#). $\square$

Proof of Theorem 1.2. By [Theorem 5.7](#), the collection is closed normalized. [Theorem 5.5](#) gives the asserted nonzero minimizer. $\square$

6. COMPACT TRAJECTORY HULLS

6.1. Trajectory hulls.

Definition 6.1 (Trajectory hull). For a bounded mild ancient solution V , its trajectory hull is

$$\mathcal{H}(V) = \overline{\{\Theta_s V : s \in \mathbb{R}\}}^{C_{\text{loc}}^\infty}.$$

6.2. Compactness of bounded hulls.

Theorem 6.2 (Compact trajectory hull). *If V is a bounded mild ancient solution, then $\mathcal{H}(V)$ is compact in the locally smooth topology and invariant under every time translation Θ_s . Every $W \in \mathcal{H}(V)$ is a bounded mild ancient solution with*

$$\|W\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}.$$

Proof. Every translate $\Theta_s V$ is a bounded mild ancient solution with the same L^∞ bound as V . Hence every sequence of translates has a locally smoothly convergent subsequence by [Theorem 3.5](#). If $(W_n) \subset \mathcal{H}(V)$, choose translates $\Theta_{s_n} V$ with $d(W_n, \Theta_{s_n} V) < 1/n$. A subsequence of $\Theta_{s_n} V$ converges locally smoothly, and the corresponding subsequence of W_n has the same limit. Thus $\mathcal{H}(V)$ is sequentially compact. Since the locally smooth topology is metrizable, it is compact.

The L^∞ estimate and the mild equation pass to elements of the hull by local smooth convergence and [Theorem 3.4](#). For invariance, note that Θ_t maps the orbit of V onto itself. If $W_n \rightarrow W$ locally smoothly, then $\Theta_t W_n \rightarrow \Theta_t W$ locally smoothly because time translation only shifts the compact time interval on which the seminorms are evaluated. Hence $\Theta_t \mathcal{H}(V) = \mathcal{H}(V)$. $\square$

Corollary 6.3 (Uniform tightness on hulls). *If V is uniformly L^3 -tight with bound L and modulus ω , then every $W \in \mathcal{H}(V)$ satisfies*

$$\sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} \leq L$$

and

$$\sup_{\tau \in \mathbb{R}} \int_{|y| > R} |W(y, \tau)|^3 dy \leq \omega(R), \quad R \geq 1.$$

The same estimates hold uniformly on every compact invariant subset $K \subset \mathcal{H}(V)$.

Proof. Every translate of V has the same L^3 bound and tightness modulus. Applying [Theorem 4.2](#) to any sequence of translates converging locally smoothly to $W \in \mathcal{H}(V)$ gives the displayed estimates for W . Taking the supremum over $W \in K \subset \mathcal{H}(V)$ gives the final assertion. $\square$

6.3. Compact invariant sets.

Definition 6.4 (Compact invariant set). A nonempty compact subset K of the locally smooth space of bounded mild ancient solutions is invariant if

$$\Theta_t K = K \quad \text{for every } t \in \mathbb{R}.$$

It is minimal if it contains no proper nonempty compact invariant subset.

Lemma 6.5 (Continuity of the translation action). *Let K be a compact invariant set. For each $t \in \mathbb{R}$, $\Theta_t : K \rightarrow K$ is a homeomorphism, and the action*

$$\mathbb{R} \times K \ni (t, W) \mapsto \Theta_t W \in K$$

is jointly continuous.

Proof. For fixed t , convergence $W_n \rightarrow W$ locally smoothly implies

$$\|\Theta_t W_n - \Theta_t W\|_{C^m(B_R \times [-T, T])} = \|W_n - W\|_{C^m(B_R \times [-T+t, T+t])} \rightarrow 0.$$

Thus Θ_t is continuous, and its inverse is Θ_{-t} . If $t_n \rightarrow t$ and $W_n \rightarrow W$, then on any fixed compact cylinder the difference

$$\Theta_{t_n} W_n - \Theta_t W$$

is bounded by a term involving $W_n - W$ on a slightly larger compact cylinder and a term involving the uniform continuity of finitely many derivatives of W on that larger cylinder. Both terms tend to zero, proving joint continuity. $\square$

6.4. Minimal compact invariant subsets.

Proposition 6.6 (Minimal compact invariant subsets). *Let K be a nonempty compact invariant set. Then K contains a nonempty compact minimal invariant subset.*

Proof. Let $\mathcal{F}$ be the family of nonempty compact invariant subsets of K , ordered by reverse inclusion. For a chain $\{K_\alpha\}$, the intersection $\bigcap_\alpha K_\alpha$ is nonempty by compactness and the finite intersection property, since the chain is totally ordered by ordinary inclusion. The intersection is compact and invariant: invariance follows from $\Theta_t K_\alpha = K_\alpha$ for all α and the fact that Θ_t is a homeomorphism. Thus every chain has an upper bound in the reverse-inclusion order. Zorn's lemma gives a maximal element for that order, equivalently a minimal compact invariant subset under ordinary inclusion. $\square$

6.5. The zero trajectory.

Lemma 6.7 (The zero trajectory). *The singleton $\{0\}$ is compact and invariant. If a minimal compact invariant set contains the zero trajectory, then it equals $\{0\}$.*

Proof. The zero solution satisfies $\Theta_t 0 = 0$ for all t . Therefore $\{0\}$ is a nonempty compact invariant subset of any invariant set containing zero. Minimality forces equality. $\square$

7. INVARIANT MEASURES AND THE COVARIANCE OBSTRUCTION

7.1. Krylov–Bogolyubov averages. The following is the standard Krylov–Bogolyubov averaging argument [7, 1].

Theorem 7.1 (Krylov–Bogolyubov averages). *Let K be a compact invariant set and $W_0 \in K$. For $T > 0$, define*

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\Theta_s W_0} ds.$$

Then every sequence $T_n \rightarrow \infty$ has a subsequence such that

$$\mu_{T_n} \rightharpoonup \mu$$

weakly as Borel probability measures on K , and every such limit μ is invariant:

$$(\Theta_t)_\# \mu = \mu \quad \text{for all } t \in \mathbb{R}.$$

If K is minimal, then every invariant Borel probability measure on K has support equal to K .

Proof. The orbit map $s \mapsto \Theta_s W_0$ is continuous by Theorem 6.5; thus μ_T is the push-forward of normalized Lebesgue measure on $[0, T]$. Since K is compact and metrizable, the space of Borel probability measures on K is weakly sequentially compact [1]. Hence μ_{T_n} has a weakly convergent subsequence.

For $F \in C(K)$ and $t > 0$,

$$\int_K F(\Theta_t W) d\mu_T(W) - \int_K F(W) d\mu_T(W) = \frac{1}{T} \int_T^{T+t} F(\Theta_s W_0) ds - \frac{1}{T} \int_0^t F(\Theta_s W_0) ds.$$

The absolute value is at most $2t\|F\|_{C(K)}/T$. Letting $T = T_{n_k} \rightarrow \infty$ gives invariance for $t > 0$; negative times follow by applying the positive-time result to $F \circ \Theta_t$.

If K is minimal and μ is invariant, then $\text{supp } \mu$ is nonempty, closed, and invariant. Indeed, if U is a neighborhood of $\Theta_t W$, then $\Theta_{-t}U$ is a neighborhood of W , and invariance shows $\mu(U) = \mu(\Theta_{-t}U) > 0$ whenever $W \in \text{supp } \mu$. Hence $\Theta_t(\text{supp } \mu) \subset \text{supp } \mu$. Applying the same argument with $-t$ gives the reverse inclusion, so $\text{supp } \mu$ is invariant. It is nonempty because $\mu(K) = 1$, and it is compact because it is closed in compact K . Minimality forces $\text{supp } \mu = K$. $\square$

7.2. Support on minimal hulls.

Corollary 7.2 (Nonzero support). *Let K be a minimal compact invariant set with $K \neq \{0\}$. Then zero is not in K , and every invariant probability measure on K has support equal to K .*

Proof. If zero were in K , [Theorem 6.7](#) would give $K = \{0\}$, contrary to the assumption. The support conclusion is the minimal-support part of [Theorem 7.1](#). $\square$

7.3. Stationary statistical identity.

Theorem 7.3 (Stationary statistical form). *Let K be a compact invariant set and μ an invariant Borel probability measure on K . Then for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,*

$$(7.1) \quad 0 = \int_K \left[\int_{\mathbb{R}^3} W(y, 0) \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) dy + \int_{\mathbb{R}^3} W_i(y, 0) W_j(y, 0) \partial_j \phi_i(y) dy \right] d\mu(W).$$

Proof. Define

$$F(W) = \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) dy.$$

Local smooth convergence makes F continuous on K : if $W_n \rightarrow W$, then $W_n(\cdot, 0) \rightarrow W(\cdot, 0)$ uniformly on the compact support of ϕ . Invariance gives

$$\int_K \frac{F(\Theta_h W) - F(W)}{h} d\mu(W) = 0 \quad (0 < |h| \leq 1).$$

Choose R with $\text{supp } \phi \subset B_R$. On $\overline{B}_R \times [-1, 1]$, compactness of K in the locally smooth topology gives

$$M_\phi = \sup_{W \in K} \|\partial_\tau W\|_{L^\infty(\overline{B}_R \times [-1, 1])} < \infty.$$

For $0 < |h| \leq 1$,

$$\frac{F(\Theta_h W) - F(W)}{h} = \int_{\mathbb{R}^3} \left(\frac{1}{h} \int_0^h \partial_\tau W(y, s) ds \right) \cdot \phi(y) dy,$$

and the absolute value is bounded by $|B_R| \|\phi\|_{L^\infty} M_\phi$, uniformly in W and h . For each fixed W , smoothness gives uniform convergence of the inner average to $\partial_\tau W(y, 0)$ on $\text{supp } \phi$. Dominated convergence on K yields

$$\int_K \int_{\mathbb{R}^3} \partial_\tau W(y, 0) \cdot \phi(y) dy d\mu(W) = 0.$$

For each W , the smooth local equation [\(1.1\)](#) at time zero, tested against solenoidal ϕ , gives

$$\begin{aligned} \int_{\mathbb{R}^3} \partial_\tau W \cdot \phi dy &= \int_{\mathbb{R}^3} W \cdot \Delta \phi dy + \int_{\mathbb{R}^3} W \cdot \phi dy + \frac{1}{2} \int_{\mathbb{R}^3} W \cdot (y \cdot \nabla \phi) dy \\ &\quad + \int_{\mathbb{R}^3} W_i W_j \partial_j \phi_i dy. \end{aligned}$$

The pressure term vanishes because $\text{div } \phi = 0$; the drift identity uses

$$\int y_j \partial_j W_i \phi_i dy = -3 \int W \cdot \phi dy - \int W \cdot (y \cdot \nabla \phi) dy.$$

The right-hand side is a continuous bounded function of $W \in K$. Averaging over K gives [\(7.1\)](#). $\square$

7.4. Barycenter and covariance.

Theorem 7.4 (Barycenter and covariance). *Let K and μ be as in [Theorem 7.3](#). Define*

$$\overline{W}(y) = \int_K W(y, 0) d\mu(W)$$

and

$$C_{ij}(y) = \int_K W_i(y, 0)W_j(y, 0) d\mu(W) - \overline{W}_i(y)\overline{W}_j(y).$$

Then $\overline{W}$ and C are C_{loc}^∞ , $\overline{W}$ is solenoidal, and for every solenoidal $\phi \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$,

$$(7.2) \quad \int_{\mathbb{R}^3} \overline{W} \cdot \left(\Delta \phi + \phi + \frac{1}{2} y \cdot \nabla \phi \right) dy + \int_{\mathbb{R}^3} \overline{W}_i \overline{W}_j \partial_j \phi_i dy + \int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i dy = 0.$$

In particular, $\overline{W}$ is a weak stationary self-similar solution against solenoidal test fields if and only if

$$\int_{\mathbb{R}^3} C_{ij} \partial_j \phi_i dy = 0$$

for every solenoidal test field ϕ .

Proof. Compactness of K in the locally smooth topology gives, for every compact ball and every derivative order, a uniform bound on $\partial_y^\alpha W(y, 0)$ for $W \in K$. Hence one may differentiate $\overline{W}$ under the integral sign. More explicitly, for every multiindex α and every y in a fixed compact ball,

$$\partial_y^\alpha \overline{W}(y) = \int_K \partial_y^\alpha W(y, 0) d\mu(W),$$

because the difference quotients are dominated by the next derivative seminorm, uniformly on K . The same argument applies to the second moment

$$M_{ij}(y) = \int_K W_i(y, 0)W_j(y, 0) d\mu(W),$$

using Leibniz' rule and the uniform bounds on derivatives of W . This gives C_{loc}^∞ regularity of $\overline{W}$, M , and $C = M - \overline{W} \otimes \overline{W}$. Since every W is divergence free, Fubini against scalar test functions gives $\text{div } \overline{W} = 0$: for $\eta \in C_c^\infty(\mathbb{R}^3)$,

$$\int_{\mathbb{R}^3} \overline{W} \cdot \nabla \eta dy = \int_K \int_{\mathbb{R}^3} W(y, 0) \cdot \nabla \eta(y) dy d\mu(W) = 0.$$

For each compactly supported test field ϕ , the maps

$$W \mapsto \int_{\mathbb{R}^3} W(y, 0) \cdot \phi(y) dy, \quad W \mapsto \int_{\mathbb{R}^3} W_i(y, 0)W_j(y, 0) \partial_j \phi_i(y) dy$$

are continuous and bounded on K , again by compactness in C_{loc}^∞ . Thus all integrals below are finite, and Fubini's theorem applies.

Applying [Theorem 7.3](#) and passing the linear term through the K -integral gives the first integral in (7.2). Passing the nonlinear term through the integral gives

$$\int_K W_i W_j d\mu = \overline{W}_i \overline{W}_j + C_{ij}.$$

Substitution yields (7.2). Comparing this identity with the pressure-free stationary weak formulation for $\overline{W}$ gives the stated equivalence with vanishing covariance contribution. $\square$

8. STRICT LYAPUNOV FUNCTIONALS AND EQUILIBRIUM SUPPORT

8.1. Stationary trajectories.

Definition 8.1 (Stationary trajectory). An element W of a compact invariant hull K is stationary if

$$\Theta_t W = W \quad \text{for every } t \in \mathbb{R}.$$

Equivalently, $W(y, \tau) = w(y)$ is independent of τ .

Lemma 8.2 (Stationary trajectories and stationary profiles). *If W is stationary, then $W(y, \tau) = w(y)$, and w solves (4.1) in distributions for some scalar distribution p .*

Proof. Stationarity means $W(y, \tau + t) = W(y, \tau)$ for all t , hence $W(y, \tau) = W(y, 0) = w(y)$. Conversely, a time-independent trajectory is fixed by every Θ_t . By Theorem 3.2, for each N there is a smooth pressure p_N on B_N such that

$$(w \cdot \nabla)w + \nabla p_N = \Delta w - \frac{1}{2}(w + y \cdot \nabla w) \quad \text{on } B_N.$$

On overlaps, the gradients of two local pressures agree, so their difference is constant on each connected overlap. More explicitly, if $M > N$, then on B_N both gradients equal the same smooth vector field

$$\Delta w - \frac{1}{2}(w + y \cdot \nabla w) - (w \cdot \nabla)w.$$

Thus $\nabla(p_M - p_N) = 0$ on B_N , and $p_M - p_N$ is constant there. Choose the additive constants in the pressures inductively so that $p_{N+1} = p_N$ on B_N . If $\varphi \in C_c^\infty(\mathbb{R}^3)$, choose N with $\text{supp } \varphi \subset B_N$ and define

$$\langle p, \varphi \rangle = \langle p_N, \varphi \rangle.$$

The compatibility on overlaps makes this independent of N . Testing the local identity on a ball containing the support of a vector test field then shows that (4.1) holds in $\mathcal{D}'(\mathbb{R}^3)$, with $\text{div } w = 0$. $\square$

8.2. The strict Lyapunov hypothesis.

Hypothesis 8.3 (Strict compact-hull Lyapunov functional). Let K be a compact invariant hull. There exist continuous functions

$$E : K \rightarrow \mathbb{R}, \quad D : K \rightarrow [0, \infty),$$

such that for every $W \in K$ and every $t \geq 0$,

$$(8.1) \quad E(\Theta_t W) - E(W) = - \int_0^t D(\Theta_s W) ds,$$

and

$$(8.2) \quad D(W) = 0 \iff W \text{ is stationary.}$$

Remark 8.4. The construction of such a Lyapunov pair (E, D) is not asserted in this paper. The role of the paper is to prove the consequences of this hypothesis for uniformly L^3 -tight compact ancient dynamics.

8.3. Invariant measures annihilate dissipation.

Lemma 8.5 (Invariant measures annihilate Lyapunov dissipation). *Let K satisfy [Theorem 8.3](#), and let μ be an invariant Borel probability measure on K . Then*

$$\int_K D(W) d\mu(W) = 0.$$

Proof. Fix $t > 0$. Integrating [\(8.1\)](#) with respect to the invariant measure μ gives

$$0 = - \int_K \int_0^t D(\Theta_s W) ds d\mu(W).$$

The integrand is continuous on the compact product $[0, t] \times K$, so Fubini applies. Invariance of μ with the continuous test function D gives

$$\int_K \int_0^t D(\Theta_s W) ds d\mu(W) = \int_0^t \int_K D(W) d\mu(W) ds = t \int_K D(W) d\mu(W).$$

Since $t > 0$, the claim follows. $\square$

8.4. Equilibrium support.

Theorem 8.6 (Equilibrium support). *Let K be a compact invariant hull satisfying [Theorem 8.3](#). Then every invariant Borel probability measure on K is supported on stationary trajectories.*

Proof. By [Theorem 8.5](#),

$$\int_K D d\mu = 0.$$

Since $D \geq 0$ is continuous, the sets

$$K_\varepsilon = \{W \in K : D(W) \geq \varepsilon\}$$

have zero μ -measure for every $\varepsilon > 0$. Therefore $\mu(K \setminus \{D = 0\}) = 0$. By [\(8.2\)](#), $\{D = 0\}$ is exactly the set of stationary trajectories. Since D is continuous, this stationary set is closed. $\square$

8.5. Minimal compact hulls are stationary.

Corollary 8.7 (Minimal compact hulls are stationary). *Let K be a minimal compact invariant hull satisfying [Theorem 8.3](#). Then every element of K is stationary. Consequently K consists of a single stationary trajectory.*

Proof. By [Theorem 7.1](#), K carries an invariant Borel probability measure μ , and minimality gives $\text{supp } \mu = K$. By [Theorem 8.6](#), μ is supported on the closed set of stationary trajectories. Hence $K = \text{supp } \mu$ is contained in that set. If $W \in K$ is stationary, the singleton $\{W\}$ is compact and invariant. Minimality of K forces $K = \{W\}$. $\square$

8.6. Compact tight rigidity.

Theorem 8.8 (Rigidity of compact tight minimal dynamics). *Let K be a minimal compact invariant hull satisfying [Theorem 8.3](#). Assume*

$$\sup_{W \in K} \sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then every element of K is identically zero.

Proof. By [Theorem 8.7](#), $K = \{W\}$ with $W(y, \tau) = w(y)$. By [Theorem 8.2](#), w solves [\(4.1\)](#). The assumed uniform L^3 bound gives $w \in L^3(\mathbb{R}^3)$. Therefore [Theorem 4.4](#) applies and gives $w = 0$. $\square$

9. CONDITIONAL TIGHT-BRANCH LIOUVILLE CRITERION

9.1. Recurrence and generated Lyapunov properties. From a minimal element V_* , the full trajectory hull $\mathcal{H}(V_*)$ is compact and invariant. It is not a consequence of compactness alone that this hull contains a nonzero minimal compact invariant subset; the local limit dynamics may admit the zero singleton as a minimal invariant subset. We therefore isolate the required recurrence input.

Definition 9.1 (The recurrence property $\text{RC}(A)$). Let A be a uniformly tight normalized collection. We say that A satisfies $\text{RC}(A)$ if every L^∞ -minimal element $V_* \in A$ has a trajectory hull

$$\mathcal{H}(V_*)$$

containing a compact minimal invariant subset K with $K \neq \{0\}$.

Definition 9.2 (The generated Lyapunov property $\text{LY}(A)$). Let A be a uniformly tight normalized collection satisfying $\text{RC}(A)$. We say that A satisfies $\text{LY}(A)$ if every compact minimal invariant subset K supplied by $\text{RC}(A)$ satisfies the strict compact-hull Lyapunov hypothesis [Theorem 8.3](#).

9.2. The conditional tight Liouville theorem.

Theorem 9.3 (No nonempty tight class under compact recurrence and Lyapunov). *No nonempty uniformly tight normalized collection A can satisfy both $\text{RC}(A)$ and $\text{LY}(A)$. Equivalently, if A is a uniformly tight normalized collection satisfying $\text{RC}(A)$ and $\text{LY}(A)$, then $A = \emptyset$.*

Proof. Suppose A is nonempty. By [Theorems 5.5](#) and [5.7](#), choose an L^∞ -minimal element $V_* \in A$.

By $\text{RC}(A)$, the trajectory hull $\mathcal{H}(V_*)$ contains a compact minimal invariant subset K with $K \neq \{0\}$. Since V_* is uniformly L^3 -tight, [Theorem 6.3](#) gives

$$\sup_{W \in K} \sup_{\tau \in \mathbb{R}} \|W(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

By $\text{LY}(A)$, K satisfies the strict Lyapunov hypothesis. Hence [Theorem 8.8](#) implies that every element of K is identically zero, so $K = \{0\}$. This contradicts $K \neq \{0\}$. Therefore no such nonempty A exists. $\square$

Proof of [Theorem 1.3](#). This is [Theorem 9.3](#). $\square$

9.3. Interface with the companion residual criterion.

Definition 9.4 (Tight-class Liouville input). The tight-class Liouville input is the assertion that no nonzero normalized Seregin ancient limit generated by an admissible Type I source is uniformly L^3 -tight.

Lemma 9.5 (Vanishing velocity removes pressure normalization). *Let (V, Π) be a smooth centered ancient solution of [\(1.1\)](#). If $V \equiv 0$, then $\Pi(y, \tau) = b(\tau)$ for some function of time. Consequently any pressure-normalized local nonvanishing condition of the form*

$$\iint_Q (|V|^3 + |\Pi - b(\tau)|^{3/2}) \, dy \, d\tau > 0$$

for every admissible pressure gauge $b(\tau)$ forces $V \not\equiv 0$.

Proof. If $V \equiv 0$, the centered equation gives $\nabla \Pi = 0$ in distributions. Thus Π is independent of y on each time slice, so $\Pi(y, \tau) = b(\tau)$. Choosing this pressure gauge makes both terms in the displayed pressure-normalized integral vanish on every compact cylinder Q . $\square$

Lemma 9.6 (From normalized Seregin nonvanishing to local velocity normalization). *Let V be a nonzero smooth bounded normalized Seregin ancient limit. Then there exist $R_0 > 0$ and $\eta_0 > 0$ such that*

$$\sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |V(y, \tau)|^3 dy \geq \eta_0.$$

Proof. Since $V \not\equiv 0$, there is (y_0, τ_0) with $V(y_0, \tau_0) \neq 0$. By continuity, $|V| \geq c > 0$ on a small spatial ball centered at y_0 at time τ_0 . Choose R_0 large enough to contain that ball. Then

$$\int_{B_{R_0}} |V(y, \tau_0)|^3 dy > 0.$$

Taking η_0 to be half of this positive value gives the claim. $\square$

Corollary 9.7 (Tight-class input for the residual criterion). *Assume that every uniformly L^3 -tight normalized Seregin collection generated by admissible Type I sources satisfies RC(A) and LY(A). Then Theorem 9.4 holds.*

Proof. Suppose, contrary to Theorem 9.4, that the companion residual-criterion paper produces a nonzero normalized Seregin ancient limit V in the uniformly L^3 -tight class. If one starts instead from a pressure-normalized nonvanishing condition in that construction, then Theorem 9.5 justifies the conclusion that $V \not\equiv 0$. Choose M , L , and a tightness modulus from V , for instance

$$M = \|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})}, \quad L = \sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^3(\mathbb{R}^3)},$$

and

$$\omega(R) = \sup_{\tau \in \mathbb{R}} \int_{|y| > R} |V(y, \tau)|^3 dy.$$

By Theorem 9.6, choose R_0, η_0 giving local velocity nonvanishing. Let

$$A_{V, R_0, \eta_0} = \left\{ W \in \mathcal{H}(V) : \sup_{\tau \in \mathbb{R}} \int_{B_{R_0}} |W(y, \tau)|^3 dy \geq \eta_0 \right\}.$$

This is the normalized compact-hull collection generated by V . It is nonempty because $V \in A_{V, R_0, \eta_0}$. It is invariant because $\mathcal{H}(V)$ is invariant and the normalization uses a supremum over all τ . If $W_n \in A_{V, R_0, \eta_0}$, $W_n \rightarrow W$ locally smoothly, and W retains the same (R_0, η_0) -nonvanishing condition, then $W \in \mathcal{H}(V)$ because $\mathcal{H}(V)$ is closed, and hence $W \in A_{V, R_0, \eta_0}$. Thus A_{V, R_0, η_0} has exactly the closedness property required in Theorem 5.6.

By Theorems 6.2 and 6.3, every element of $\mathcal{H}(V)$, hence every element of A_{V, R_0, η_0} , satisfies the same L^∞ bound, L^3 bound, and tightness modulus. Thus A_{V, R_0, η_0} is a uniformly tight normalized Seregin collection with the displayed data.

The assumed recurrence and generated Lyapunov properties apply to this collection, so Theorem 9.3 gives $A_{V, R_0, \eta_0} = \emptyset$. This contradicts $V \in A_{V, R_0, \eta_0}$. Hence no such nonzero uniformly tight Seregin limit exists under the stated assumptions, which is precisely Theorem 9.4. $\square$

The results above reduce the uniformly L^3 -tight ancient branch to two compact-dynamical inputs: nonzero compact recurrence of L^∞ -minimal elements and the strict Lyapunov equilibrium property on the resulting minimal compact hulls. Under these inputs the tight branch is empty, because the Lyapunov principle forces minimal compact dynamics to be stationary and

the stationary L^3 self-similar theorem forces the stationary profile to vanish. Thus, within the present framework, the tight-class Liouville input used in the companion residual criterion is reduced to the compact recurrence and strict Lyapunov assertions isolated here.

REFERENCES

- [1] P. Billingsley, *Convergence of Probability Measures*, second ed., Wiley, New York, 1999.
- [2] G. Duran Ballester, *Type I blow-up limits and a residual-class criterion for the three-dimensional Navier–Stokes equations*, preprint, 2026.
- [3] L. Escauriaza, G. Seregin, and V. Šverák, $L_{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness, *Russian Math. Surveys* 58 (2003), 211–250.
- [4] E. B. Fabes, B. F. Jones, and N. M. Riviere, *The initial value problem for the Navier–Stokes equations with data in L^p* , *Arch. Rational Mech. Anal.* 45 (1972), 222–240.
- [5] Y. Giga and T. Miyakawa, *Solutions in L_r of the Navier–Stokes initial value problem*, *Arch. Rational Mech. Anal.* 89 (1985), 267–281.
- [6] T. Kato, *Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions*, *Math. Z.* 187 (1984), 471–480.
- [7] N. Kryloff and N. Bogoliouboff, *La theorie generale de la mesure dans son application a l’etude des systemes dynamiques de la mecanique non lineaire*, *Ann. of Math.* (2) 38 (1937), 65–113.
- [8] O. A. Ladyzhenskaya, V. A. Solonnikov, and N. N. Ural’tseva, *Linear and Quasilinear Equations of Parabolic Type*, American Mathematical Society, Providence, 1968.
- [9] J. Nečas, M. Růžička, and V. Šverák, *On Leray’s self-similar solutions of the Navier–Stokes equations*, *Acta Math.* 176 (1996), 283–294.
- [10] H. Sohr, *The Navier–Stokes Equations: An Elementary Functional Analytic Approach*, Birkhauser, Basel, 2001.
- [11] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, Princeton, 1970.

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Email address: guillem@fragile.tech